WORKED SOLUTION

STEP II 1989, Question 2

Pure • Power Series Coefficients

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q2

The key idea

A trigonometric identity that links a function at x to the same function at $2x$ turns into a link between power-series coefficients. Replacing x by $2x$ multiplies the coefficient of x^n by 2^n, so once both sides are written as power series, we can compare coefficients and read the relation straight off. The useful trick is to multiply through by x first, so that every term is a function with a genuine power series at 0.

Solution

Throughout, x is small and non-zero, so every series below converges (all the functions involved are well behaved for $0 < |x| < \frac{\pi}{4}$, say). Two convergent power series that agree on an interval around 0 have equal coefficients, so we may compare coefficients of x^n.

The relation between a_{n-1} and b_n

First we check $(*)$:

$$\cot x - \tan x = \frac{\cos^2 x - \sin^2 x}{\sin x \cos x} = \frac{\cos 2x}{\frac{1}{2}\sin 2x} = 2\cot 2x.$$

Multiply by x:

$$x\cot x - x\tan x = (2x)\cot(2x).$$

Now write each term as a series.

- $x\cot x = 1 + \sum_{n\geq 1} b_n x^n$.
- $x\tan x = \sum_{m\geq 0} a_m x^{m+1} = \sum_{n\geq 1} a_{n-1}x^n$, putting $n = m+1$.
- Since $2x$ is small whenever x is, we may replace x by $2x$ in the series for $x\cot x$:

$$(2x)\cot(2x) = 1 + \sum_{n\geq 1} b_n (2x)^n = 1 + \sum_{n\geq 1} 2^n b_n x^n.$$

So

$$1 + \sum_{n\geq 1} (b_n - a_{n-1}) x^n = 1 + \sum_{n\geq 1} 2^n b_n x^n.$$

The constant terms agree, and comparing coefficients of x^n for $n \geq 1$ gives $b_n - a_{n-1} = 2^n b_n$, that is,

$$a_{n-1} = (1 - 2^n)\, b_n \qquad (n \geq 1),$$

as required.

Ask yourself Why multiply by x before expanding, rather than working with $\cot x$ itself?

$\cot x$ blows up at 0, so it has no power series there; the question only gives us a series for $x \cot x$. Multiplying the whole identity by x makes every term of the form "function with a power series", and the cotangent term on the right becomes $(2x)\cot(2x)$, which is the same function evaluated at $2x$. That is exactly the shape we need.

The coefficients c_n

The companion identity is the sum instead of the difference:

$$\cot x + \tan x = \frac{\cos^2 x + \sin^2 x}{\sin x \cos x} = \frac{1}{\frac{1}{2}\sin 2x} = 2\operatorname{cosec} 2x.$$

Multiplying by x again,

$$x\cot x + x\tan x = (2x)\operatorname{cosec}(2x).$$

Using the same expansions as before, and the series for $x \operatorname{cosec} x$ with x replaced by $2x$,

$$1 + \sum_{n\geq 1} (b_n + a_{n-1}) x^n = 1 + \sum_{n\geq 1} 2^n c_n x^n,$$

so for $n \geq 1$

$$2^n c_n = b_n + a_{n-1}.$$

Now use the first result to eliminate b_n. Since $n \geq 1$, $1 - 2^n \neq 0$, so $b_n = \dfrac{a_{n-1}}{1 - 2^n}$, and

$$2^n c_n = a_{n-1}\left(\frac{1}{1-2^n} + 1\right) = a_{n-1}\frac{2 - 2^n}{1 - 2^n} = a_{n-1}\frac{2\,(2^{n-1} - 1)}{2^n - 1}.$$

Dividing by 2^n, and writing $2/2^n = 1/2^{n-1}$,

$$c_n = \frac{2^{n-1} - 1}{2^n - 1} \cdot \frac{1}{2^{n-1}}\, a_{n-1} \qquad (n \geq 1),$$

which is the required result.

Checks

From $\tan x = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \cdots$ we have $a_1 = 1$, $a_3 = \frac{1}{3}$. The first result predicts $b_2 = a_1/(1-4) = -\frac{1}{3}$ and $b_4 = a_3/(1-16) = -\frac{1}{45}$, matching $x\cot x = 1 - \frac{1}{3}x^2 - \frac{1}{45}x^4 - \cdots$. The second predicts $c_2 = \frac{1}{3}\cdot\frac{1}{2}\cdot 1 = \frac{1}{6}$ and $c_4 = \frac{7}{15}\cdot\frac{1}{8}\cdot\frac{1}{3} = \frac{7}{360}$, matching $x\operatorname{cosec} x = 1 + \frac{1}{6}x^2 + \frac{7}{360}x^4 + \cdots$. Note also $n = 1$: $a_0 = 0$ and $b_1 = c_1 = 0$, consistent with both formulae. We checked both relations exactly (with fractions) for all $n \le 23$ by dividing the series for sin and cos, and checked the series against $\tan 0.3 \approx 0.309336$ and $0.3\operatorname{cosec} 0.3 \approx 1.015159$.

Common slips

- Forgetting the shift of index: $x\tan x$ has coefficient a_{n-1}, not a_n, at x^n.
- Substituting $2x$ for x in $x\cot x$ but writing $2x\cot x$ or $x\cot 2x$; the term in the identity must be exactly $(2x)\cot(2x)$.
- Not noting that $1 - 2^n \neq 0$ before dividing by it.

Beyond STEP

The coefficients here are, up to simple factors, the Bernoulli numbers: $x\cot x = \sum_{k\ge 0}(-1)^k \frac{2^{2k}B_{2k}}{(2k)!}x^{2k}$, and the question's relations give the tangent and cosecant series in terms of them. The fact that we may equate coefficients of two power series that agree near 0 (the identity theorem for power series) is proved in a first analysis course. For more, see Abbott, *Understanding Analysis*, on power series, and Needham, *Visual Complex Analysis*, where these expansions arise naturally from complex functions.

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